

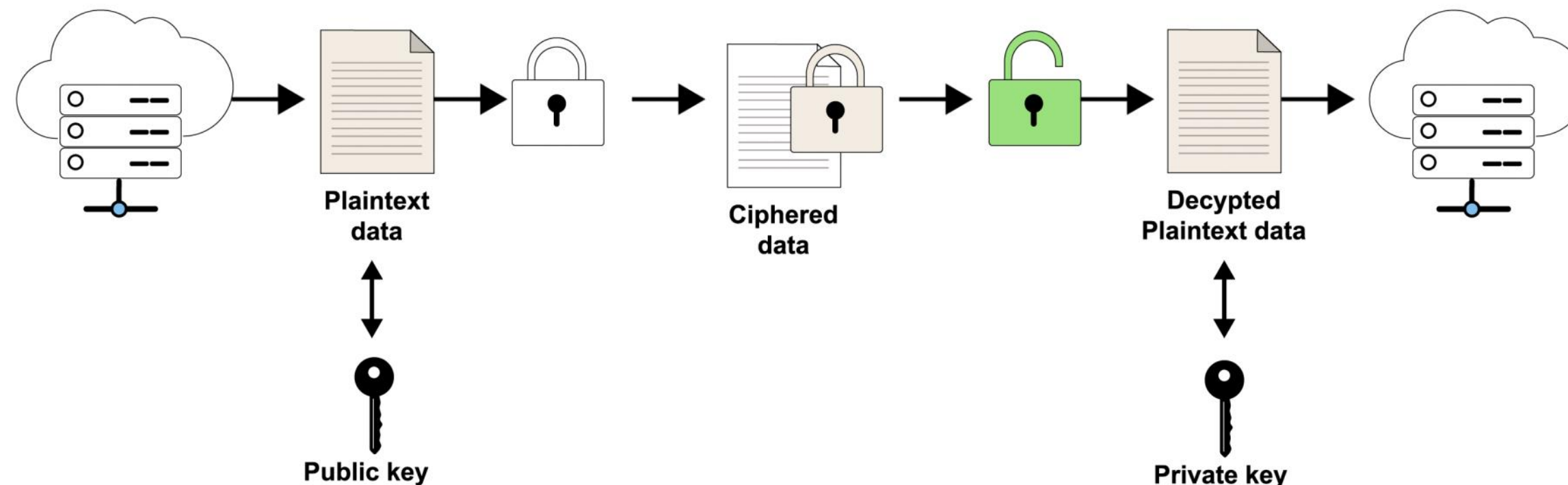
Asymmetric cryptography

Learning objectives

- **Define key pair encryption**
- **Explain how public and private keys work**
- **Explore use cases like secure email and web security**

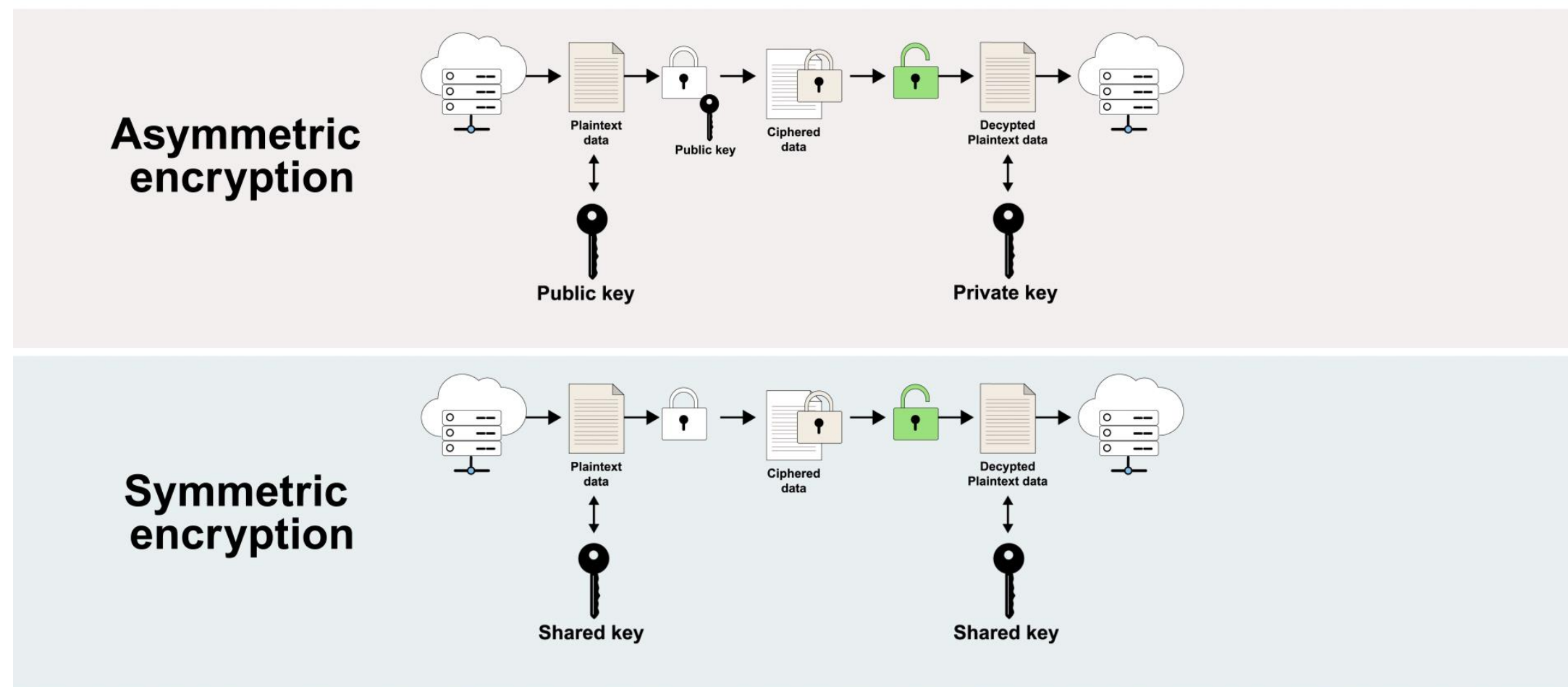
What is asymmetric cryptography?

- Uses two keys: public and private
- One encrypts, the other decrypts
- Keys work together to protect data



How asymmetric differs from symmetric encryption

- No need for sender and receiver to share a single secret
- Keys are different but linked
- More secure for open networks



Key concepts and terms

- **Public key: encrypt or verify**
- **Private key: decrypt or sign**
- **Key pair: two linked keys that work together**
- **Encryption: protects information**
- **Digital signatures: prove identity and message integrity using a private key**

How public key encryption works

- **Sender uses receiver's public key to encrypt**
- **Only receiver's private key can decrypt**
- **Used for privacy and secure messages**

Digital signatures

- **Sender signs using their private key**
- **Receiver verifies with public key**
- **Ensures authenticity and message integrity**

Use cases of asymmetric cryptography

- **Secure websites and email**
- **Blockchain and cryptocurrencies**
- **Software licensing and virtual private networks (VPNs)**

Advantages and limitations

- **Secure without pre-shared keys**
- **Supports authentication**
- **Slower than symmetric methods**

Common algorithms

- **Rivest-Shamir-Adleman (RSA): widely used for encryption and signing**
- **Elliptic Curve Cryptography (ECC): secure with smaller key sizes**
- **Diffie Hellman: for secure key exchange**

Summary

- **Asymmetric cryptography secures communication**
- **Used for encryption, identity and trust**

Next

- **Combining symmetric and asymmetric in hybrid models**